

Subject Name: Logistics and Supply Chain Management

Session No: 04

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BASICS OF LOGISTICS & SUPPLY CHAIN

Transportation, Unitising & Logistics Costing

Primary & Secondary Transportation • Unitising • Logistics Costing • Cost as a Ratio of Value Added/Sales • Integrated Supply Chain Performance

Department of Business Management



What We Will Cover

01 Primary & Secondary Transportation

The two distinct legs that move goods from factory to final customer

02 Unitising

Combining goods into standard load units for efficient handling

03 Logistics Costing

Identifying, classifying and measuring the true cost of logistics

04 Cost as a Ratio of Value Added / Sales

Benchmarking logistics performance in relative, not just absolute, terms

05 Integration for Supply Chain Performance

Bringing every activity together into one high-performing system



SECTION 01

Primary & Secondary Transportation

Every product travels in two very different legs before it reaches a customer — each with its own economics and design rules

What is Primary Transportation?

Primary transportation is the movement of bulk goods from the point of production (factory or plant) to storage or distribution points — typically a central or regional warehouse. It moves large volumes over longer distances, in as few, large shipments as possible, to minimise cost per unit.



Origin: Plant / Factory

Starts at the point of manufacture or a supplier's facility.



Destination: DC / Warehouse

Ends at a central or regional storage and consolidation point.



Bulk, Full-Load Shipments

Uses full truckloads, rail wagons or containers wherever possible.



Long-Haul, Low Frequency

Fewer, larger shipments over longer distances to minimise cost.

What is Secondary Transportation?

Secondary transportation is the onward movement of goods from a warehouse or distribution centre to the final point of sale — retail stores, dealers or the end customer's doorstep. It handles smaller, more frequent shipments across shorter distances, prioritising speed and delivery precision over cost per unit.



Origin: DC / Warehouse

Starts at a regional or local distribution point.



Destination: Store / Customer

Ends at the retail outlet or the customer's doorstep.



Smaller, Mixed Shipments

Multiple SKUs and part-loads tailored to each destination's order.



Short-Haul, High Frequency

Frequent, time-critical deliveries prioritising speed and accuracy.

Primary vs. Secondary Transportation

Primary Transportation

- Plant / factory to warehouse or DC
- Full truckload, rail or container shipments
- Long distance, low frequency
- Cost efficiency is the dominant objective
- Typically planned days to weeks in advance

Secondary Transportation

- Warehouse / DC to retail store or customer
- Part-load, multi-drop delivery routes
- Short distance, high frequency
- Speed and delivery precision dominate
- Often planned and adjusted daily

Designing Primary Transportation Well



Load Consolidation

Combine orders across products and destinations to fill trucks, rail wagons or containers completely.



Mode Selection

Rail and sea are often used for primary legs where volumes and lead times allow, since they are far cheaper per unit than road.



Network Flow Planning

Shipment frequency and routing are planned around production schedules, not customer orders.



Cost per Unit Focus

The key metric is landed cost per case or per tonne-kilometre, not delivery speed.

Designing Secondary Transportation Well



Route Optimisation

Multi-drop delivery routes are sequenced to minimise total distance and delivery time windows.



Fleet Right-Sizing

Smaller vehicles suited to urban access and part-loads replace the large vehicles used upstream.



Time-Window Commitments

Delivery slots are promised and tracked — lateness has a direct, visible cost to customer service.



Real-Time Tracking

GPS and proof-of-delivery systems are essential given the volume and frequency of stops.

Where the Optimisation Opportunities Lie

Optimising Primary Legs

- Shift eligible volumes from road to rail or sea
- Maximise vehicle and container fill rates
- Consolidate shipments across product lines
- Negotiate long-term freight contracts for stable lanes

Optimising Secondary Legs

- Cluster deliveries by geography and time window
- Use route-optimisation software for daily planning
- Balance dedicated fleet vs. on-demand courier capacity
- Track cost-to-serve by customer and channel

Example: The Two-Leg Journey of a Packaged Snack

A snack-food company moves product from its factory in Nashik to a retail shelf in Nagpur through two distinct transportation legs.

Primary

A full 32-tonne truck moves 20,000 cases from the Nashik factory to the Nagpur regional DC, twice a week

Secondary

Smaller vans deliver mixed pallets of 50–200 cases from the DC to 60+ retail stores across the city, daily

Result

Primary cost per case is 70% lower than secondary — exactly why the two legs are planned so differently

LESSON



Treating both legs with the same transportation strategy would either waste capacity on the primary leg or fail delivery precision on the secondary leg — the two must be designed separately.



SECTION 02

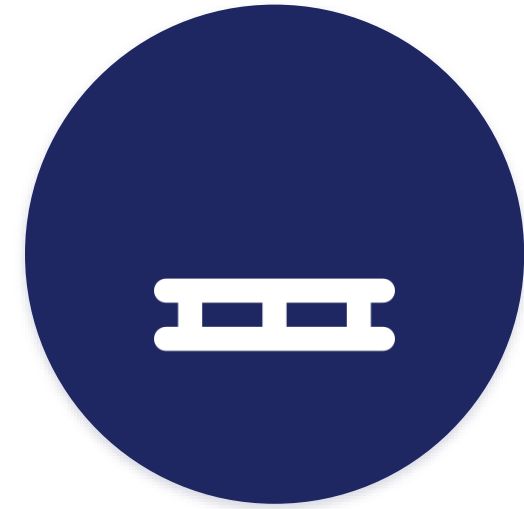
Unitising

Combining individual items into standard load units is one of the simplest ideas in logistics — and one of the highest-leverage

What is Unitising?

Unitising is the practice of grouping individual items, cartons or packages into a single, standard-sized load unit — such as a pallet, container or roll cage — so that goods can be handled, stored and transported as one unit instead of many separate pieces.

Important slide



EXAMPLE



Instead of loading 500 individual cartons of biscuits by hand, a warehouse stacks them onto 10 standard pallets — one forklift movement replaces fifty manual carries.

Common Types of Unit Loads



Pallets

The most common unit load — wooden or plastic platforms holding stacked cartons, moved by forklift.



Roll Cages / Dollies

Wheeled cages ideal for retail deliveries where goods are rolled straight onto the shop floor.



Shipping Containers

Standardised steel containers (20ft/40ft) that unitise cargo for sea, rail and road movement.



Cartons & Cases

The smallest standard unit — individual items grouped into a single retail-ready case.



Intermediate Bulk Containers

Large containers for liquids, powders or granular material moved and stored as one unit.



Slip Sheets

A thin, low-cost alternative to pallets for lighter loads, saving space and weight.

Why Unitising Pays for Itself



Faster Handling

One pallet move replaces dozens of individual carton movements, cutting labour time sharply at every touchpoint.



Reduced Damage & Loss

Goods secured as a single unit are far less likely to be dropped, crushed or pilfered in transit.



Better Space Utilisation

Standard unit sizes stack and store predictably, maximising warehouse cube and truck-load efficiency.



Lower Handling Cost

Fewer touches per shipment directly reduces labour and equipment cost per unit shipped.



Easier Tracking

One barcode or RFID tag per unit load simplifies inventory counting and shipment tracking.

Unit Load Devices & Standards

Unitising only delivers its full benefit when load sizes are standardised across the whole supply chain — otherwise pallets don't fit trucks, and trucks don't fit warehouse racking.



ISO Pallet Standard

1200mm × 1000mm (Euro-standard) or 1200mm × 800mm — sized to fit standard truck and container widths.



ISO Shipping Containers

20-foot and 40-foot standard containers, sized for interchangeable use across ships, rail and trucks.



Warehouse Racking Standards

Rack beam widths and heights are designed around standard pallet footprints.

Unitising Across Different Transport Modes

Primary (Long-Haul) Transport

- Full pallets and containers are the norm
- Maximises fill rate on trucks, rail and ships
- Minimises handling between origin and DC
- Standard sizes allow intermodal transfer

Secondary (Last-Mile) Transport

- Roll cages and smaller unit loads dominate
- Mixed-SKU units built to match store orders
- Faster unloading at multi-stop delivery routes
- Reusable units reduce packaging waste

Example: Unitising in a Beverage Distribution Centre

A beverage company's distribution centre handles over 40,000 cases a day. Before unitising, workers manually stacked cartons onto trucks.

4hrs → 45min

Time to load a full truck after switching to standard pallets

30%

Reduction in transit damage claims within the first quarter

18%

More cases fit per truck due to consistent, stackable pallet heights

LESSON



The switch to standard pallets required no change to the product itself — the entire gain came from handling goods as unit loads instead of individual cartons.



SECTION 03

Logistics Costing

You cannot manage what you cannot measure — logistics costing gives visibility into where money actually goes

What is Logistics Costing?

Logistics costing is the process of identifying, classifying and measuring all the costs incurred in moving, storing and managing goods through the supply chain — giving managers the visibility needed to make informed trade-offs between cost and service.



Visibility

Reveals exactly where logistics spend goes, function by function.



Trade-off Analysis

Enables informed cost-vs-service decisions across the network.



Performance Tracking

Provides the baseline to measure improvement over time.



Pricing & Budgeting

Feeds accurate product costing and departmental budgets.

The Major Components of Logistics Cost



Transportation Cost

Freight, fuel, tolls and carrier charges across primary and secondary legs.



Warehousing Cost

Rent, utilities, labour and equipment for storage and handling facilities.



Inventory Carrying Cost

Cost of capital, insurance, obsolescence and shrinkage on stock held.



Order Processing & Admin

Order management, documentation, customer service and IT systems.



Packaging Cost

Materials and labour for protective and unitised packaging.



Loss, Damage & Returns

The cost of goods lost, damaged in transit, or returned by customers.

Direct vs. Indirect Logistics Costs

Direct Costs

- Clearly traceable to a specific shipment or order
- Freight charges, fuel surcharges, packaging materials
- Vary directly and predictably with volume
- Easier to allocate accurately to a product or customer

Indirect Costs

- Shared across many shipments, products or customers
- Warehouse rent, IT systems, management overhead
- Do not vary directly and immediately with volume
- Require an allocation method to assign fairly

Activity-Based Costing (ABC) in Logistics

Traditional costing spreads overhead evenly across all products or customers. Activity-based costing instead traces cost to the specific activities — picking, packing, delivering — that each product or customer actually consumes.

1

Identify Activities

e.g. picking, packing, loading, delivering

**2**

Assign Cost Drivers

e.g. number of picks, pallets moved, stops made

**3**

Trace Costs

Link activity costs to specific orders or customers

**4**

Reveal True Cost-to-Serve

Some customers cost far more to serve than others

Total Logistics Cost: Looking Beyond Freight

The single biggest costing mistake is judging logistics decisions on freight rate alone. Total logistics cost captures every cost a decision touches — often reversing what looks like the "cheaper" choice.

Total Logistics Cost = Transportation + Warehousing + Inventory Carrying + Order Processing + Packaging + Loss/Damage

Every relevant cost bucket, added together — not just the freight invoice

EXAMPLE



A cheaper overseas supplier with a lower unit price can still raise total logistics cost once longer transit time, higher safety stock, and greater damage risk are added in.

Approaches to Logistics Costing



Standard Costing

Uses predetermined cost rates per unit or activity, compared against actuals to spot variances.



Activity-Based Costing (ABC)

Traces cost to the specific activities each order or customer actually consumes.



Mission / Segment Costing

Costs a specific distribution channel, customer segment or geographic route end-to-end.



Total Cost of Ownership (TCO)

Captures every lifecycle cost of a sourcing or network decision, not just the purchase price.

Example: Logistics Cost Breakdown for a Retail Chain

A mid-size retail chain analyses its full logistics cost structure for the year:

42%

Transportation — the single largest cost component, split across primary and secondary legs

28%

Warehousing — rent, labour and handling equipment across the DC network

18%

Inventory carrying cost — capital, insurance and obsolescence on stock held

INSIGHT



With transportation identified as the largest lever, the company launches a route-optimisation project targeting secondary transportation — the highest-frequency, most controllable cost within that 42%.



SECTION 04

Cost as a Ratio of Value Added or Sales

A rupee figure alone tells you little — expressing logistics cost as a ratio makes it comparable across products, years and companies

Why Express Logistics Cost as a Ratio?

An absolute logistics cost figure cannot tell you whether performance is good or bad — a larger company will always spend more in rupee terms. Expressing cost as a ratio of sales or value added makes performance comparable across time, products, business units and competitors.



EXAMPLE



Company A spends ₹50 crore on logistics, Company B spends ₹20 crore. Company A looks worse — until you learn Company A's sales are ₹1,000 crore (5% ratio) and Company B's are ₹250 crore (8% ratio).

Logistics Cost as a Percentage of Sales

$$\text{Logistics Cost Ratio} = (\text{Total Logistics Cost} / \text{Net Sales Revenue}) \times 100$$

The most widely used logistics performance benchmark

- Simple to calculate and universally understood across finance and operations teams
- Allows direct comparison against industry benchmarks and prior-year performance
- A falling ratio over time signals improving logistics efficiency relative to revenue growth
- Can be distorted by pricing changes — a price cut alone worsens the ratio even if logistics is unchanged

Logistics Cost as a Percentage of Value Added

Value Added = Sales Revenue – Cost of Purchased Materials & Services

What the company itself contributes, after stripping out bought-in inputs

Logistics cost is then expressed against this value-added base rather than total sales:

Logistics Cost as % of Value Added = (Total Logistics Cost / Value Added) × 100

A better measure when comparing companies with very different levels of outsourced input

% of Sales vs. % of Value Added: When to Use Which

% of Sales

- Simplest to calculate — uses figures every firm reports
- Best for tracking one company's trend over time
- Distorted by how much a company outsources or buys in
- The standard external benchmarking metric

% of Value Added

- Strips out the effect of bought-in materials and services
- Best for comparing companies with different make-vs-buy mixes
- More complex — requires a value-added calculation
- The preferred metric for cross-industry comparison

Typical Logistics Cost Ratios by Industry

4–6%

Consumer packaged goods (FMCG) — high volume, dense distribution networks

8–12%

Industrial & heavy manufacturing — bulky, lower-value-density goods

15–20%+

E-commerce & fragmented retail — high last-mile cost per order

Ratios vary widely by product weight-to-value density, distribution complexity and geography — a single company should be benchmarked against its own industry, not a generic average.

CAUTION



A very low ratio is not always good news — it can also signal under-investment in service levels, leading to lost sales that never show up in the logistics cost line at all.

Example: Calculating and Interpreting the Ratio

A consumer electronics company reports the following for the year:

Total Logistics Cost = ₹80 crore | Net Sales = ₹1,600 crore

Reported figures for the financial year

Logistics Cost Ratio = $(80 / 1,600) \times 100 = 5.0\%$

Compared with an industry benchmark of 6–8% for this category

INTERPRETATION



At 5.0%, the company is running more efficiently than its industry peers — a strong signal to protect and study, not automatically a target for further cost-cutting that could hurt service.



SECTION 05

Integration of All Activities

Transportation, unitising and costing only deliver their full value when every activity is planned and managed as one connected system

What is Supply Chain Integration?

Supply chain integration is the alignment and coordination of all activities — procurement, production, inventory, transportation, warehousing and customer service — across functions and organisations, so they operate toward one shared performance goal rather than optimising in isolation.



Cross-Functional

Sales, operations, finance and logistics plan together, not in silos.



Cross-Organisational

Suppliers and customers are treated as partners in the same system.



End-to-End Flow

Material, information and financial flows move as one connected process.

The Three Dimensions of Integration



Internal Integration

Aligning sales, marketing, production, logistics and finance around one shared plan and one set of numbers within the company.



Upstream (Supplier) Integration

Sharing forecasts, capacity plans and quality data with suppliers so they can plan ahead rather than react.



Downstream (Customer) Integration

Sharing demand and inventory visibility with distributors and retailers to reduce the bullwhip effect.

Common Barriers to Integration



Functional Silos

Departments are measured on separate goals that can directly conflict with each other.



Disconnected Systems

Different IT platforms for sales, production and logistics prevent shared visibility.



Trust & Data Sharing

Companies are reluctant to share forecasts or costs with suppliers and customers.



Cultural Resistance

Long-standing habits and incentive structures resist cross-functional collaboration.



Conflicting Metrics

A logistics team rewarded for low cost may work against a sales team rewarded for service.



Geographic & Organisational Distance

Multiple business units or countries make coordination structurally harder.

What Enables Real Integration



Shared Information Systems

A common ERP, forecasting and tracking platform that every function and partner can see.



Integrated Planning Processes

S&OP-style forums that force cross-functional trade-offs to be made together, not separately.



Aligned Performance Metrics

Shared KPIs (like total logistics cost as % of sales) that make collaboration, not conflict, the rational choice.



Collaborative Relationships

Long-term supplier and customer partnerships built on shared data and shared risk.



Leadership Commitment

Executive sponsorship that resolves cross-functional conflicts and sustains the change.

The Performance Impact of Integration

↓ Cost

Lower total logistics cost as a % of sales through eliminated duplication and waste

↑ Service

Higher fill rates and on-time delivery from shared visibility across the chain

↓ Inventory

Less safety stock needed when demand and supply signals are shared, not guessed

Every topic in this course connects here: the transportation legs, the unitising decisions, and the cost ratios only reach their full potential when planned as one integrated system, not five separate ones.

Example: Integration in Practice at a Consumer Goods Company

A consumer goods company integrates five previously separate functions into one coordinated system:

- Sales shares point-of-sale data directly with production planning, instead of monthly order forecasts alone
- Logistics and warehousing standardise on the same unit-load (pallet) sizes as procurement's supplier contracts
- A single logistics-cost-as-%-of-sales metric is reviewed monthly by every function, not just finance
- Key suppliers get visibility into the company's production schedule to plan their own capacity

Read this slide

RESULT



Within a year, total logistics cost as a percentage of sales falls from 7.2% to 5.8%, while on-time delivery improves from 89% to 96% — proof that integration improves cost and service together, not one at the expense of the other.



KEY TAKEAWAYS

From Individual Moves to One System

- Primary and secondary transportation are distinct legs, each needing its own mode, load size and planning approach
- Unitising — pallets, containers, roll cages — cuts handling cost, damage and time at every touchpoint
- Logistics costing gives visibility into true cost, well beyond the freight invoice alone
- Expressing cost as a ratio of sales or value added makes performance genuinely comparable
- None of it reaches full potential without integrating every activity into one connected system

Thank You